

PROGRAM 23

AIM

To implement a Prolog-based diet recommendation system that suggests suitable food items based on the disease selected by the user.

ALGORITHM

1. Start the program.
2. Define diseases and their recommended food items as facts.
3. Define rules to obtain diet recommendations for a given disease.
4. Accept the disease name from the user.
5. Search the knowledge base for the corresponding diet.
6. Display the recommended food items.
7. If no matching disease is found, display an appropriate message.
8. Stop the program.

PROGRAM

```
diet(diabetes, [vegetables, oats, whole_grains, nuts]).  
diet(hypertension, [fruits, vegetables, oats, low_salt_food]).  
diet(anemia, [spinach, beans, eggs, iron_rich_food]).  
diet(obesity, [vegetables, fruits, whole_grains, salads]).  
diet(gastritis, [oats, banana, rice, boiled_vegetables]).
```

```
suggest_diet(Disease) :-
```

```
    diet(Disease, Foods),  
    write('Recommended Diet: '),  
    write(Foods).
```

OUTPUT

```
35 ?- consult(['c:/Users/Hema Neeпа/OneDrive/Documents/Diet_Recommendation_System.pl']).  
true.
```

```
36 ?- suggest_diet(diabetes).  
Recommended Diet: [vegetables,oats,whole_grains,nuts]  
true.
```

```
37 ?- suggest_diet(anemia).  
Recommended Diet: [spinach,beans,eggs,iron_rich_food]  
true.
```

```
38 ?- suggest_diet(obesity).  
Recommended Diet: [vegetables,fruits,whole_grains,salads]  
true.
```

RESULT

The Prolog-based diet recommendation system was successfully implemented, and suitable food recommendations were obtained based on the selected disease.